
One Nation, One Material Code

Resolving duplicate and functionally equivalent material codes across Central Public Sector Enterprises by matching extracted specifications rather than text, and issuing a single Common National Material Code while every original CPSE code is retained and mapped.

PROBLEM STATEMENT	SIH26099
TITLE	AI-Driven Standardization and Harmonization of Material Codes Across CPSEs
MINISTRY	Ministry of Petroleum & Natural Gas
DEPARTMENT	Chennai Petroleum Corporation Limited (CPCL)
THEME	Smart Automation · Software
TEAM	Overwatch
DOCUMENT	Technical reference — architecture, method and measured results
DATE	31 August 2026

01 The problem

Central Public Sector Enterprises operating in oil and gas, power, steel, mining and heavy engineering each maintain an independent material master — the catalogue of every pipe, valve, bearing, gasket and fastener the organisation buys and stocks. The same physical item is assigned a different material code, a different description, a different unit of measurement and a different classification in each enterprise, and is frequently duplicated within a single enterprise as well.

The problem statement names six consequences. Each is addressed by a specific part of the system described in this document.

STATED IN THE PROBLEM STATEMENT	ADDRESSED BY
Duplication of material masters	Blocking, scoring and clustering → one national code per physical item
Inconsistent descriptions	Canonicaliser → noun-modifier standard description
Difficulty identifying equivalent materials	Specification-level matching with brand ignored
Fragmented procurement data	Purchase orders aggregated per national code
Higher inventory levels	Rationalisation of redundant codes
Limited collaborative procurement	Demand aggregation with federated price benchmarking

Why the descriptions diverge

Three causes compound, and the third is the largest. Attributing the problem to any single one of them is inaccurate.

- 1. Field length in the ERP.** In SAP, material descriptions are stored in table `MAKT`, field `MAKTX`, which holds a maximum of 40 characters. `MAKT` is the text table of `MARA`; the description field does not exist in `MARA` itself. Forty characters is not enough for a complete specification, so cataloguers abbreviate.
- 2. Migration from older systems.** Data loaded into current ERP instances arrived from predecessor systems with even shorter description fields, and the truncation is preserved.
- 3. No enforced cataloguing standard.** Where no naming convention is mandated and audited, each storekeeper develops a private shorthand. This is the dominant cause, and it is the reason the same item diverges even within one enterprise on one ERP.

VERIFIED

MAKT-MAKTX is defined as a 40-character material description field, and MAKT is the text table for MARA. Confirmed against the published SAP data dictionary.

02 Sources of material data

Material master data does not exist as a public dataset. It sits inside each enterprise's ERP. Four public sources establish what real material records look like, what attributes procurement actually distinguishes on, and which classification standard to adopt. Each was checked directly.

Government e Marketplace — gem.gov.in

GeM is the Government of India's national public procurement portal, used by central and state ministries, departments, PSUs and autonomous bodies. It publishes over 10,000 product categories, including the engineering categories this system covers — pumps, valves, pipes and fittings, engineering tools and safety equipment. Category listings carry structured technical parameters, and the marketplace hosts buyer-uploaded technical specification documents for items such as API 6D valves, control valves, HDPE pipes and pipeline ball valves.

DOMAIN CONCEPT ADOPTED FROM GEM

GeM defines **Golden Parameters** as those specifications which have maximum impact on the technical aspects of a product, may directly or indirectly impact its cost, and enable a buyer to choose the appropriate product. This is precisely the distinction this system encodes as **hard blocker fields** in Section 07 — the attributes on which a mismatch makes two items non-interchangeable regardless of how similar everything else is. The concept is not invented here; it is already the operating vocabulary of Indian public procurement.

Central Public Procurement Portal — eprocure.gov.in

CPPP is the portal through which all central government organisations publish tender enquiries, corrigenda and award-of-contract details. Financial bids are submitted on a standardised Bill of Quantities spreadsheet whose template structure may not be altered. Published BOQ line items are the closest public analogue to a material master row: a free-text item description, a unit of measurement and a quantity, written by a procurement officer under the same abbreviation pressures described in Section 01. BOQ item text is therefore the reference for how real description variation actually looks, rather than how it might be imagined to look.

UNSPSC — classification standard

The United Nations Standard Products and Services Code is a four-level hierarchy — segment, family, class, commodity. The system classifies to class level and embeds the class in every national code it issues. Using an international standard rather than a locally invented taxonomy matters for a national system: no CPSE is asked to adopt a scheme created for this project, and the resulting codes remain interoperable with any procurement system that already speaks UNSPSC. Every class used is listed and verified in Section 09.

SAP MM data model

The material master tables define the shape of what an enterprise would export. **MARA** holds general material data keyed by **MATNR**; **MAKT** holds descriptions per language, keyed by **MATNR** and **SPRAS**, with **MAKTX** as the 40-character short text. **MARC** holds plant-level data and **MBEW** valuation. The ingestion format mirrors this structure so an export from a live system requires no reshaping.

Sources verified 31 Aug 2026 — gem.gov.in (categories, Golden Parameters, technical specification documents) · eprocure.gov.in/cppp (BOQ structure) · UNSPSC published class listing · SAP data dictionary (**MAKT-MAKTX**, **MARA**).

03 The central technical problem

This is entity resolution over short, heavily abbreviated, domain-specific text, with no labelled training data, where a match means **functional equivalence** rather than textual similarity, at a scale that forbids all-pairs comparison.

PROPERTY OF THE DATA	CONSEQUENCE FOR THE DESIGN
Descriptions are 8–15 words	Weak signal; text alone is insufficient
Heavy abbreviation — M.S. , SS316 , SCH40 , NB	Out-of-vocabulary; a normalisation layer is required
No training labels exist	Unsupervised methods only
Equivalence is functional, not textual	Two records can be interchangeable while sharing almost no words
Masters run to millions of rows	Blocking is mandatory; all-pairs comparison is excluded
Error costs are asymmetric	A wrong merge is far more costly than a missed duplicate

Text similarity orders these two pairs incorrectly

Both pairs below are drawn from the evaluation set. Similarity is cosine distance between sentence embeddings produced by BAAI/bge-small-en-v1.5 , 384 dimensions, vectors L2-normalised.

PAIR A — different words, SAME physical item

CPCL 100001445 SKF 6205-2RS DEEP GROOVE BALL BEARING

IOCL MAT-01248 FAG 6205-2RSR BEARING, 25X52X15MM, SEALED

text similarity0.833

6205 is an ISO bearing designation: 25 mm bore, 52 mm outer diameter, 15 mm width, for any manufacturer.

PAIR B — near-identical words, DIFFERENT physical items

ONGC 5001035 GASKET SPIRAL WOUND SS316 4IN 150#

BPCL BR-01454 GASKET SPIRAL WOUND SS304 4IN 150#

text similarity0.989

SS316 contains molybdenum; SS304 does not. In corrosive service the wrong grade pits and leaks.

THE DESIGN CONSEQUENCE

The pair that must never merge scores **higher** than the pair that should. The ordering is inverted, and an inverted ordering cannot be corrected by moving a threshold: at 0.80 both merge, at 0.90 the bearing is missed and the gasket still merges, at 0.99 both fail. No setting produces the correct outcome for both. The system therefore does not tune the threshold — it changes what is measured, and matches on extracted specifications rather than on text.

04 System architecture

Nine stages run in sequence. Each writes a file that the next stage reads, so any stage can be re-run independently and every intermediate result is inspectable. Data is separated into three folders under a strict rule: inputs are read-only, outputs are regenerable, and the ground-truth answer key is read by evaluation alone and never by any matching component.

Source systems — one per CPSE, never centralised

ONGC · IOCL · CPCL · BPCL · GAIL
SAP MARA / MAKT / MARC / MBEW

Ingest

material master export
+ purchase history

01 Normalise

expand abbreviations
canonicalise UoM · fix casing

02 Extract specifications

LLM → typed attributes per category
per-field confidence · regex fallback

03 Embed

BAAI/bge-small-en-v1.5 · 384 dims
L2-normalised vectors

04 Block

category gate + key attribute
+ top-k vector neighbours

05 Score

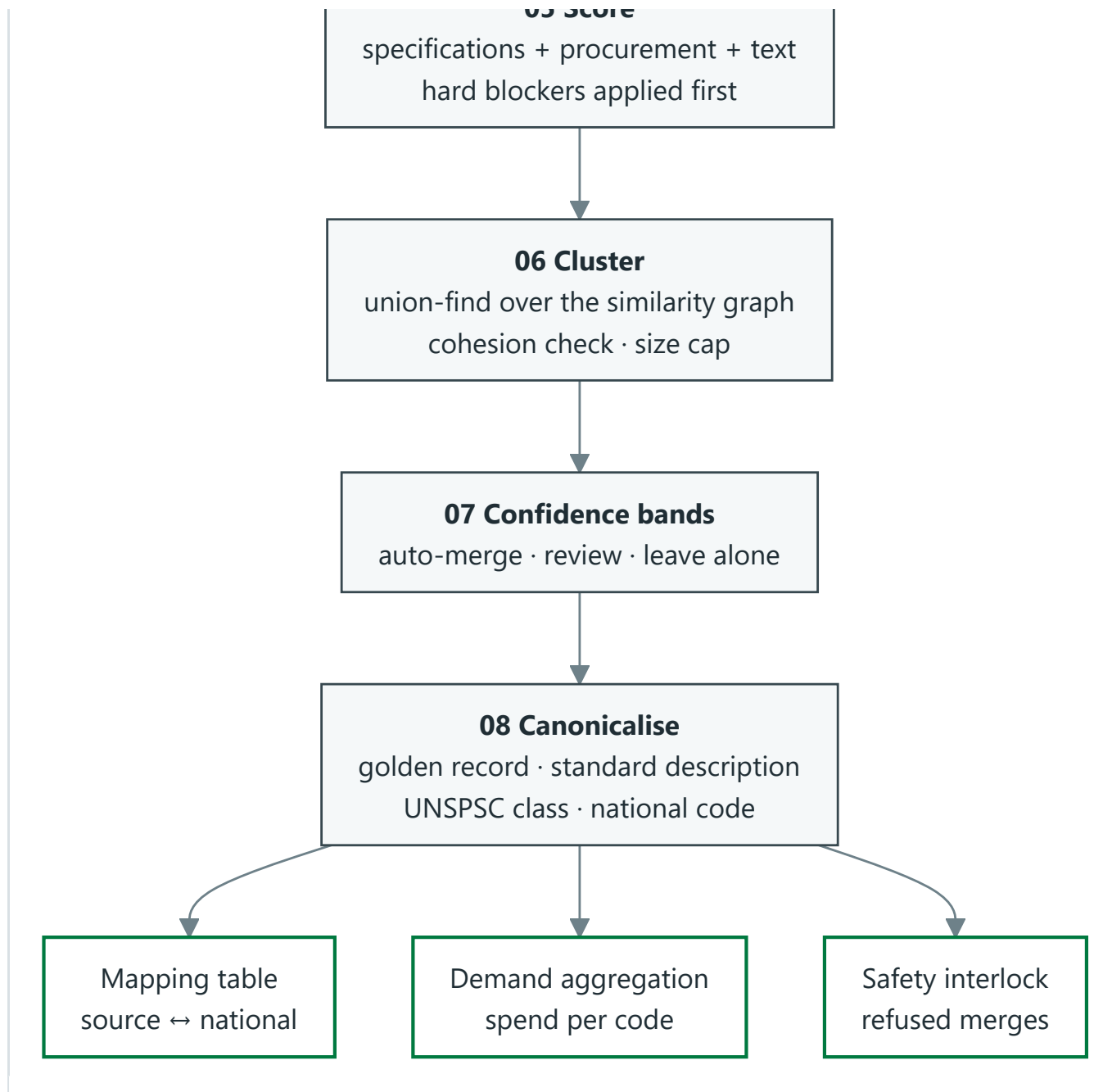


Figure 1. End-to-end pipeline. The three outputs on the right are the deliverables: what maps to what, where money is being left on the table, and what the system refused to merge.

Where the language model sits, and where it does not

The model is called **once per record, at ingestion, to extract attributes**. It is never called on a pair of records and it never decides whether two records match. Matching is deterministic arithmetic over already-extracted attributes. Two properties follow from this and both are requirements rather than optimisations.

- **Auditability.** Every merge decision can be replayed and explained from stored numbers. A hallucinated merge is structurally impossible because the model is not in the decision path.
- **Cost and latency.** One million records require one million extraction calls, executed once and stored. Calling a model per candidate pair would require tens of millions of calls on *every* run.

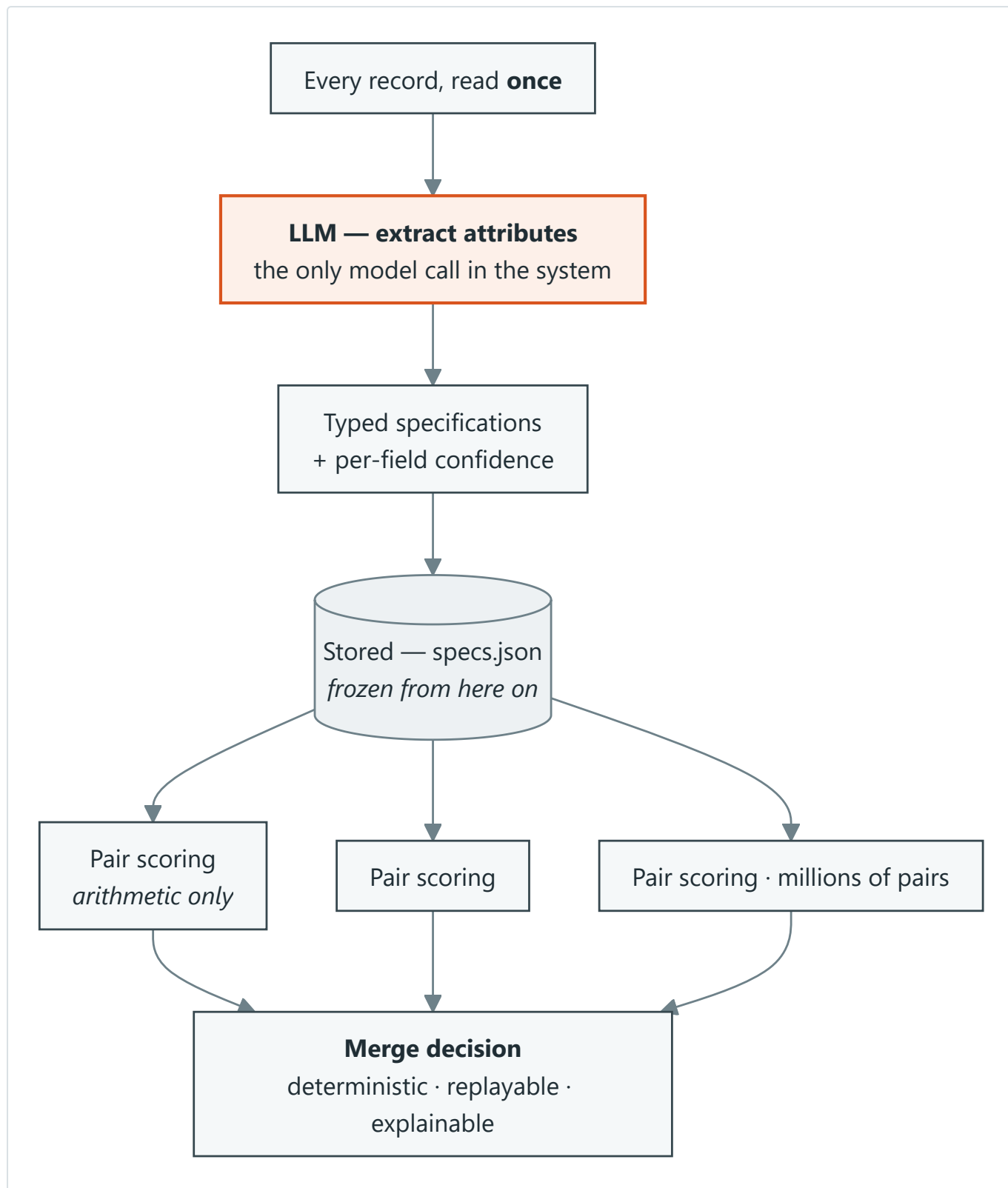


Figure 2. The model appears once, on the left, and its output is frozen. Everything downstream of `specs.json` is arithmetic, which is what makes the merge decision auditable.

05 Normalisation and unit harmonisation

The problem statement names units of measurement explicitly in its background. UoM divergence is handled as a first-class concern rather than as a side effect of text cleaning, because a unit mismatch silently corrupts price comparison: a price per piece and a price per box of one hundred are not comparable, and nothing in the description says which is which.

The evaluation set contains ten distinct unit strings representing three real units.

CANONICAL UNIT	VARIANTS PRESENT IN THE DATA	VARIANTS
EA — each	EA , EACH , NO , NOS , PC , PCS	6
M — metre	M , METER , MTR , MTS	4
2 canonical units		10

Normalisation also expands domain abbreviations before extraction runs. Dotted forms are the difficult case: once punctuation is stripped, C.S. becomes c s and no longer matches a dictionary entry for cs , so the normaliser tests both the spaced and the compacted form of every candidate. Dimensional forms vary independently of the words around them — 25X52X15MM , 25 x 52 x 15 mm and 25*52*15 M.M. are the same measurement — and are rewritten to a single form before any comparison.

06 Specification extraction

Each record is assigned a material category, and each category defines a typed schema. The extractor returns values only for fields in that schema, with a confidence value per field. Nine categories are defined.

CATEGORY	REPRESENTATIVE TYPED FIELDS
bearing	sub_type, iso_designation, bore_mm, od_mm, width_mm, seal_type, brand
gasket	gasket_type, material_grade, nominal_size_in, pressure_class
valve	valve_type, nominal_size_in, pressure_class, body_material, end_connection
pipe	pipe_type, material_grade, nominal_size_in, schedule
fastener	fastener_type, thread, length_mm, material_grade
flange	flange_type, face_type, nominal_size_in, pressure_class, material_grade
rotating	seal_type, shaft_mm, face_material, elastomer
instrument	instrument_type, range, connection, accuracy
electrical	equipment_type, rating_hp, rpm, voltage, enclosure

Derived values carry lower confidence

The bearing pair in Section 03 illustrates the mechanism. One record states 25X52X15MM ; the other states only 6205 . Because 6205 is an ISO designation with fixed dimensions, the extractor returns bore 25, outer diameter 52 and width 15 for both records — but marks them at lower confidence on the record where they were *derived* from the designation rather than *read* from the text. The distinction is preserved so that a reviewer can see which values were inferred.

Extraction is cached and resumable

Results are stored keyed by record identifier, so a re-run extracts only records not already present. A regex fallback handles every category without a model call, which keeps the pipeline operable without network access. On the evaluation set, extraction returned typed specifications for all 100 records.

07 Blocking and scoring

Blocking

Comparing every record against every other is quadratic and is excluded at any realistic scale. Candidate pairs are generated by a category gate, a key attribute match and top-k nearest neighbours in vector space. On the evaluation set, 4,376 pairs were eliminated by the category gate alone before any scoring occurred, leaving 292 pairs to score.

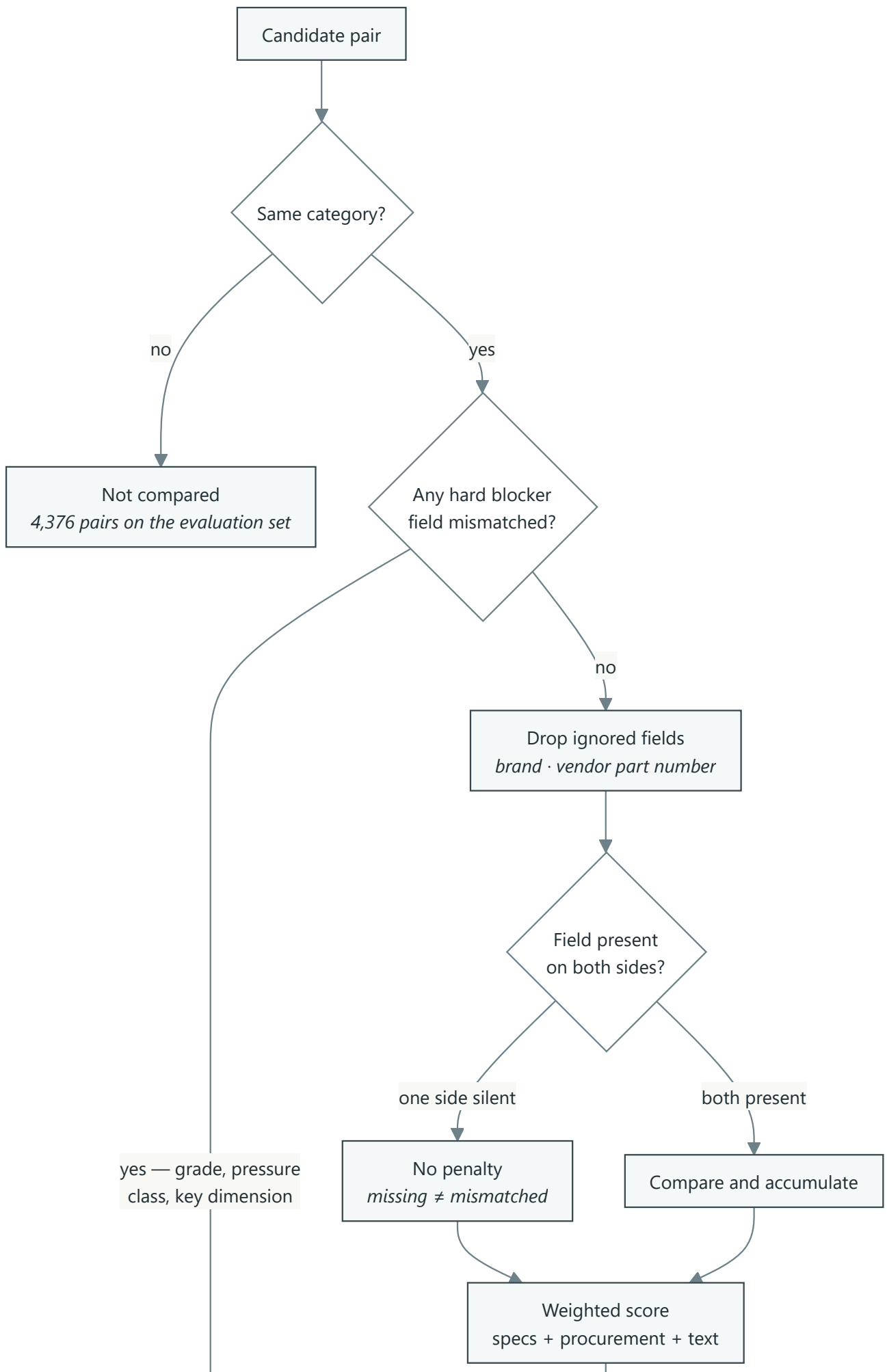
Three signals

SIGNAL	BASIS	WEIGHT
Specifications	Field-by-field agreement between extracted attribute sets	Highest. Grows with the number of fields actually verified.
Procurement behaviour	Shared vendor, comparable unit price band, order size, plant	Secondary, and independently protective
Text	Cosine similarity between sentence embeddings	Lowest. Never decisive on its own.

Specification weight is adaptive: a pair agreeing on eight verified fields carries more specification weight than a pair agreeing on two, because the evidence is stronger. Text weight falls correspondingly. This prevents the weakest signal from dragging down a pair whose specifications align completely — which is exactly the bearing case, and exactly the outcome the design exists to produce.

The procurement signal is not only additive evidence. It is also a guard against false merges: a bolt priced at ₹42 and a bolt priced at ₹4,200 cannot be the same item however similar their descriptions and specifications appear. A price ratio beyond an order of magnitude blocks the pair outright.

Three rules that matter more than the weights



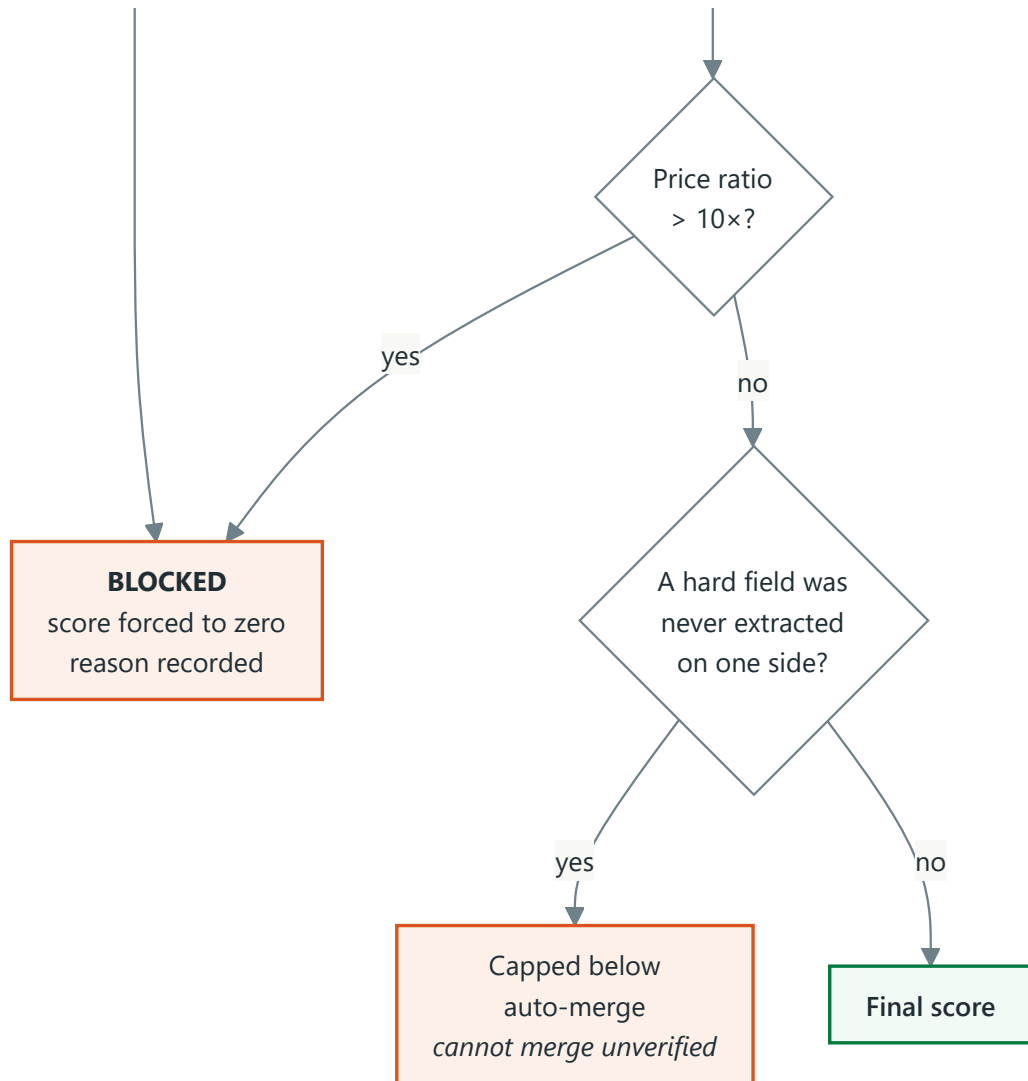


Figure 3. Scoring decision path. The hard-blocker test runs *before* any similarity is computed, so a grade mismatch cannot be outvoted by agreement everywhere else.

Hard blockers are veto fields

Material grade, pressure rating and key dimensions are veto fields. A mismatch forces the score to zero regardless of every other signal. This is the engineering property GeM calls a Golden Parameter: the attribute that determines whether one item can physically and safely replace another.

Ignored fields

Brand and vendor part number never contribute to a match. This single rule is what allows an SKF bearing and an FAG bearing carrying the same ISO designation to resolve to one item. Functional equivalence does not depend on who manufactured the part.

Missing is not mismatched

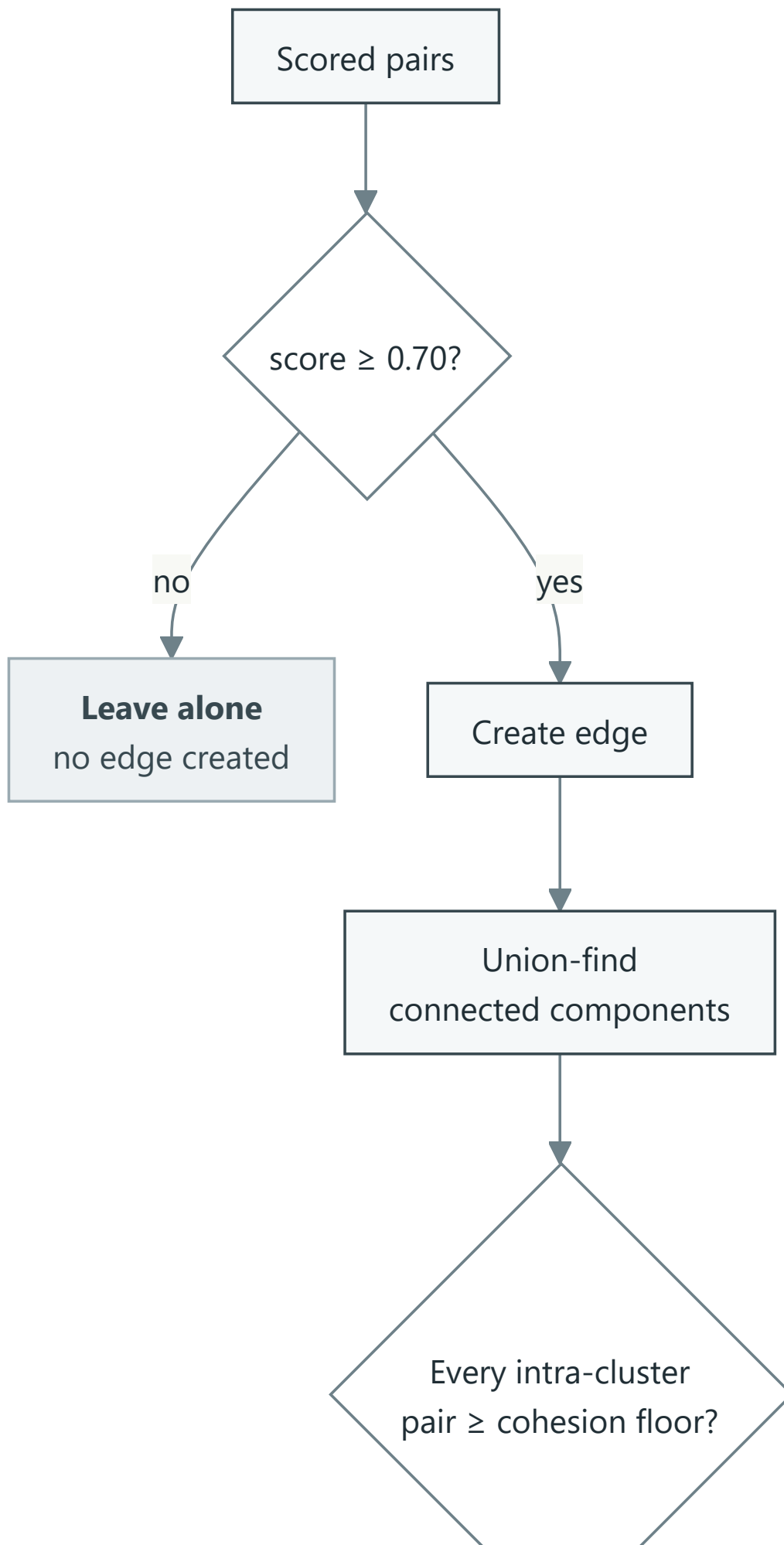
When one record is silent on a field, that is absent information, not evidence of difference, and no penalty is applied. The converse case is handled separately and strictly: if a *hard* field was never extracted from one side, the pair cannot be auto-merged, because the veto could not be evaluated. Such pairs are capped below the auto-merge threshold and routed to human review.

08 Clustering and confidence bands

Scored pairs form a graph. Connected components under union-find produce clusters, but transitive closure introduces a specific hazard: if A matches B and B matches C while A and C do not match, the component still contains all three, and a single bridging record can pull hundreds of unrelated items into one blob.

Three defences are applied.

- **High join threshold** — only pairs above the review floor create an edge.
- **Cohesion validation** — every intra-cluster pair must score above a minimum. A cluster containing a weakly attached member is split rather than accepted. Two clusters were held back on this rule in the evaluation run.
- **Size cap** — clusters beyond a maximum size are routed to review rather than merged automatically.



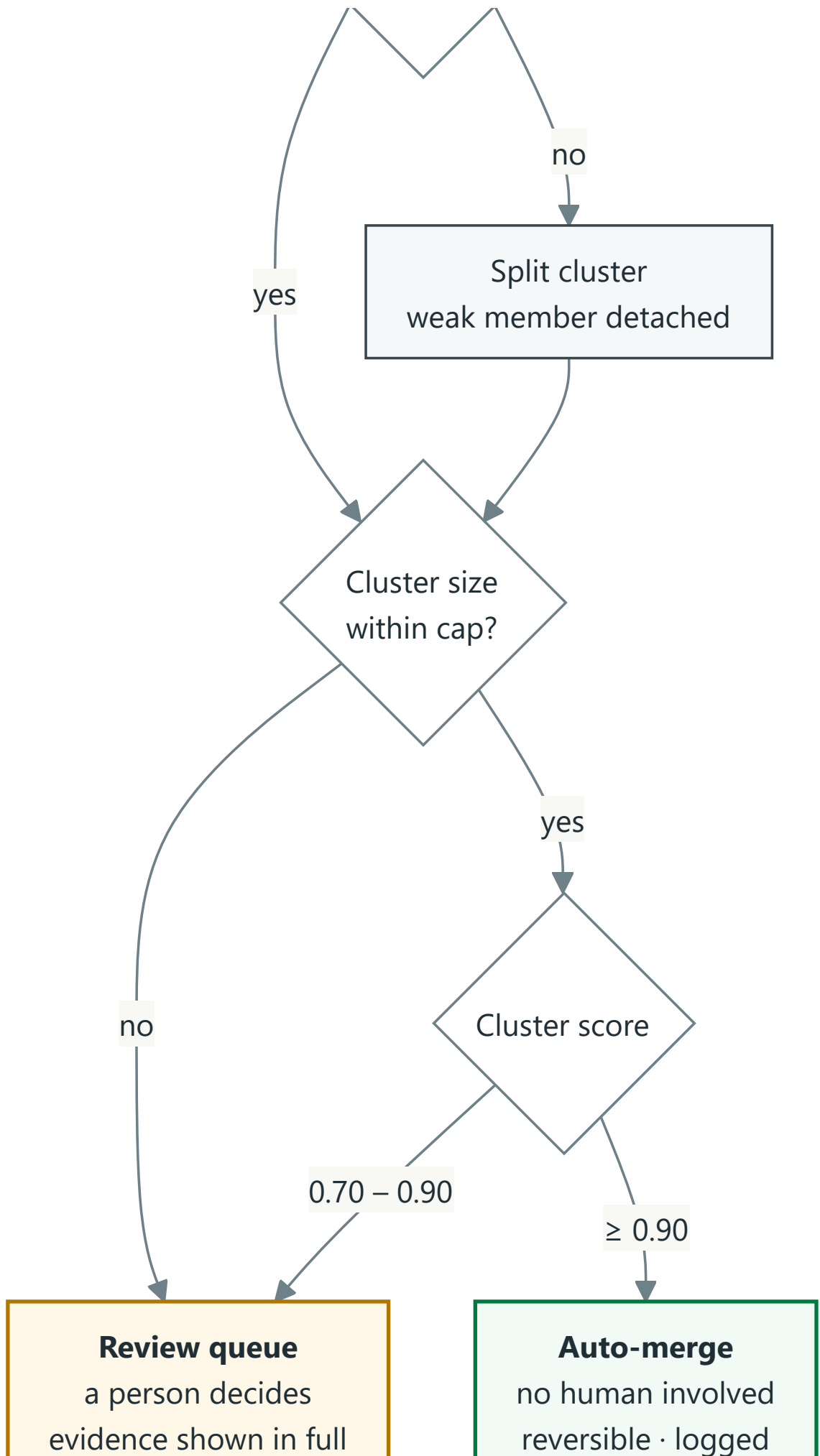




Figure 4. Clustering and confidence banding. Only the top band merges without a person; the middle band is the product surface a cataloguer works through.

Human review is the specified design

The problem statement uses the words *recommend*, *propose*, *review*, *validate* and *approve* throughout, and lists a user validation and approval workflow among its expected capabilities. The middle band exists because the specification calls for it, not as a hedge against accuracy.

Two governance rules follow. Within a single enterprise, that enterprise's own reviewer decides alone. Across two enterprises, both sides must approve before a national code is issued — no organisation decides the identity of another organisation's material. Merges are reversible: reversing one deletes mapping rows and never deletes source data, because the material table is append-only and original codes are never overwritten.

09 Canonicalisation and the national code

Each accepted cluster produces four outputs.

Golden record

The best available value for each field across all cluster members, preferring values that were read rather than derived, and higher-confidence extractions over lower.

Standard description

Noun first, then modifiers in order of significance — the format industry already uses. Once specifications are typed this is a template render, which is why it is the most direct visible evidence that extraction succeeded.

```
BEARING, DEEP GROOVE BALL, 6205, 25X52X15 MM, 2RS SEALED
GASKET, SPIRAL WOUND, SS304, 4 IN, 150#, GRAPHITE
VALVE, GATE, 6 IN, 150#, CARBON STEEL, FLANGED, API 600
FLANGE, WELD NECK, RAISED FACE, 4 IN, 150#, ASTM A105
MOTOR, 3 PHASE, 5 HP, 1440 RPM, FOOT, 415 V, IP55
BOLT, HEX, M12 X 50, SS316, IS 1364
MECHANICAL SEAL, CARTRIDGE SINGLE, 45 MM SHAFT, SIC VS CARBON, VITON
```

UNSPSC classification

Every class below was checked against the published UNSPSC class listing on 31 August 2026. The class is embedded in each national code, so an incorrect class is visible on every output row.

CATEGORY	UNSPSC CLASS	VERIFIED CLASS TITLE
bearing	31171500	Bearings
valve	40141600	Valves
fastener	31161600	Bolts
flange	40175000	Pipe weldneck flanges
pipe	40171600	Industrial pipe and piping
rotating	31411700	Mechanical seals
electrical	26101100	Electric alternating current AC motors
instrument	41111900	Indicating and recording instruments
gasket	31400000	Gaskets (family level — see below)

A CLASSIFICATION LIMIT WORTH STATING PLAINLY

UNSPSC family 31400000 divides gaskets by **manufacturing method** — molded (31401500), die cut (31401600), stamped (31401700), water jet cut (31401800) and liquid (31401900). **Spiral wound gaskets, which are standard in refinery flanged joints, have no class at all.** Assigning one to "molded gaskets" would be a wrong classification printed inside every national code issued for that item, so gaskets are classified at family level and the limitation is recorded rather than hidden. Neighbouring classes were checked and none fits: 31411800 is diaphragm seals and 31411900 is molded seals.

National code

The code carries its classification: `NMC-<UNSPSC class>-<serial>` . It sits *above* existing codes and links them, in the way Aadhaar and UPI added a layer without replacing what was underneath. No enterprise changes its own code; a CPCL storekeeper continues to type `100001445` .

NMC-31171500-000001	BEARING, DEEP GROOVE BALL, 6205, 25X52X15 MM, 2RS SEALED
	← CPCL 100001445 SKF 6205-2RS DEEP GROOVE BALL BEARING
	← IOCL MAT-01248 FAG 6205-2RSR BEARING, 25X52X15MM
	← ONGC 5001102 BEARING BALL DG 6205 2RS
	... 9 source codes across 5 CPSEs
NMC-31400000-000001	GASKET, SPIRAL WOUND, SS304, 4 IN, 150#, GRAPHITE
	10 source codes across 4 CPSEs

10 Federated operation

Commercial master-data platforms require all participating data to be centralised. No CPSE will transfer its negotiated prices and its vendor list to a shared repository, and this is a governance constraint rather than a missing feature. It is the reason a national material master has not previously existed.

The system is designed so that matching and benchmarking work without commercially sensitive values crossing an organisational boundary.

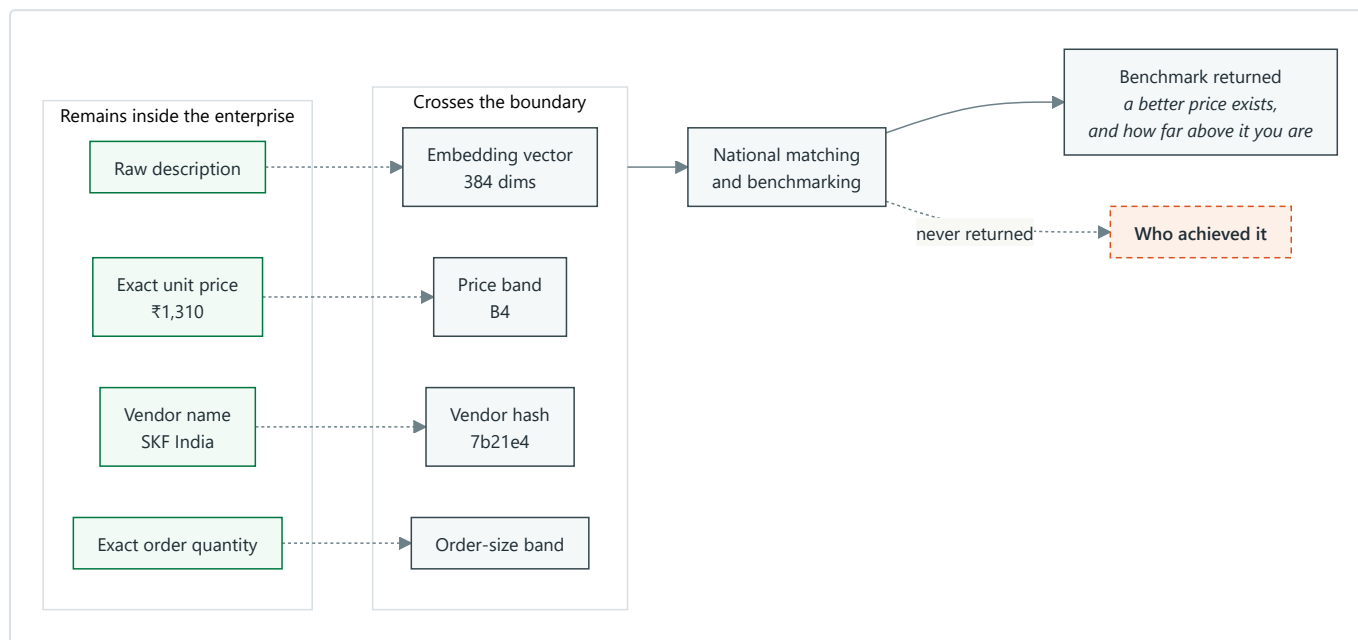


Figure 5. What crosses the boundary and what does not. Equal hashes prove two enterprises used the same vendor without naming it; band comparison detects a large price gap without revealing either price.

Benchmark, not attribution

An enterprise learns that a better price exists and how far above it their own price sits. It does not learn which organisation achieved that price. Where fewer than three enterprises appear in a cluster the benchmark is **suppressed entirely**, because with only two participants the minimum price is the other party's price. Four items were suppressed under this rule on the evaluation set.

STATED PRECISELY

This design **significantly reduces data exposure**. It is not described as cryptographically secure, and that distinction is deliberate: embeddings are partially invertible and hash-based encodings have known attacks. Secure multi-party computation is the correct answer to that stronger requirement.

11 Measured results

All figures below are measured on the evaluation set: 100 material records drawn from five CPSEs across nine categories, with 245 purchase orders covering 30 vendors and 11 plants. Precision and recall are computed against a record-level answer key that is read by the evaluation harness alone and by no matching component.

100	16	84.0%	282
RECORDS	UNIQUE ITEMS	DUPLICATION	MERGES REFUSED

Matching accuracy

MEASURE	VALUE	TARGET
Precision	100.0%	> 98%
Recall	94.2%	> 85%
F1	97.0%	—
True duplicate pairs in the answer key	310	
Correctly merged	292	
Wrongly merged	0	
Missed	18	

Auto-merge band — the safety-critical number

Pairs scoring at or above 0.90 are merged with no human involvement, so an error in this band reaches the material master directly. Reporting a single blended precision figure would hide the distinction that matters.

AUTO-MERGE BAND (≥ 0.90)	VALUE
Pairs merged without review	177
Of those, wrong	0
Precision in this band	100.0%
Deliberate near-miss pairs auto-merged	0
Pairs routed to human review (0.70 – 0.90)	115

Safety interlock — refused merges

282 candidate pairs were refused by a veto rule. This is reported as a primary output rather than a diagnostic, because in a refinery a wrongly merged grade is a leak rather than a data-quality issue.

VETO RULE	PAIRS REFUSED
Material grade mismatch — e.g. SS316 vs SS304	78
Valve type mismatch	60
ISO designation mismatch	54
Fastener type mismatch	49
Pressure class mismatch — e.g. 150# vs 300#	27
Body material mismatch	14
Total	282

Clustering

MEASURE	VALUE
Candidate pairs eliminated by the category gate	4,376
Pairs scored	292
Clusters produced	16
in the auto-merge band	12
in the review band	3
single-member	1
Clusters split on the cohesion rule	2
Clusters mixing two distinct real items	0

The two reference pairs, end to end

PAIR	TEXT	SPECS	FINAL	OUTCOME
SKF 6205-2RS / FAG 6205-2RSR	0.833	1.0	0.971	Merged
Gasket SS316 / SS304	0.989	—	0	Blocked — material grade

Demand aggregation

Purchase orders grouped by national code reveal where two or more enterprises buy the same physical item separately.

MEASURE	VALUE
Total spend analysed	₹22.61 cr
Spend on items bought by 2+ CPSEs	₹20.42 cr
Items bought by 2 or more CPSEs	15 of 16
Items eligible for a benchmark	11
Items suppressed under k-anonymity (k = 3)	4
Saving at best observed price — upper bound	₹1.79 cr
Saving at 40% capture — realistic	₹71.5 L

BOTH FIGURES ARE STATED TOGETHER, ALWAYS

The upper bound assumes every enterprise achieves the best price any of them obtained, which does not happen in practice — and any procurement officer reading this will know it. Publishing the conservative figure alongside it is what makes the optimistic one credible.

Largest single aggregation opportunity

NMC-40141600-000001 VALVE, GATE, 6 IN, 150#, CARBON STEEL, FLANGED, API 600

5 CPSEs 18 purchase orders 1,900 units ₹5.57 cr

best observed unit price ₹24,083.86

worst observed unit price ₹33,322.10

price spread 38.4%

saving at best price ₹99.3 L upper / ₹39.7 L at 40% capture

Scope of these figures

These results are measured on a 100-record evaluation set whose variation was constructed to reproduce the abbreviation, reordering, unit-format and typographic patterns observed in public procurement text. A measurement made against a constructed answer key establishes that the method behaves as designed; it does not by itself establish the figure a live material master would produce. The next measurement to make is against a hand-labelled sample of real records drawn from GeM and CPPP listings, reported alongside these numbers rather than in place of them.

12 Design rules

These constraints hold across every component and are not traded away for accuracy.

RULE	REASON
Precision over recall	A missed duplicate costs an opportunity. A wrong merge in a refinery material master can put the wrong part in a line. Thresholds are set conservatively and uncertainty is routed to a person.
Nothing is destroyed	The material table is append-only. Original codes are always retained and mapped. Reversing a merge removes mapping rows and never removes source data.
Every merge is explainable	Which signals fired, what each scored and which fields agreed are stored with the decision, so a reviewer sees the evidence rather than a number.
The model never decides a merge	It extracts attributes. Matching is deterministic scoring over extracted data, which keeps the system auditable and makes a hallucinated merge structurally impossible.
One model call per record	Never per pair. Per-record cost is linear and paid once; per-pair cost is quadratic and paid on every run.
No enterprise decides for another	Within one CPSE, that CPSE's reviewer decides alone. Across two, both must approve before a national code is issued.
The answer key is never read by the matcher	Ground truth is read by the evaluation harness alone. A matcher that has seen the answer key produces a precision figure that means nothing.

Positioning

Commercial master-data management platforms exist, and record linkage as a field dates to Fellegi and Sunter in 1969. Attribute-based matching is not new. What is claimed here is **execution and scope**: these platforms deploy inside a single enterprise, whereas this is a cross-enterprise problem whose hard parts are trust, data sharing and the issuance of one national identity; the abbreviation vocabulary is learned from the corpus rather than configured by consultants per client; and national strategic procurement data remains on sovereign infrastructure.

13 References

SOURCE	USED FOR
Problem statement SIH26099 sih.gov.in	Requirement text, key capabilities, expected impact
Government e Marketplace gem.gov.in	Product categories, technical parameters, published specification documents, the Golden Parameters concept
Central Public Procurement Portal eprocure.gov.in/cppp	Tender BOQ structure and real item-description variation
UNSPSC published class listing	Every classification code in Section 09, verified class by class
SAP data dictionary MARA • MAKT • MAKTX • MARC • MBEW	Material master structure and the 40-character description constraint
Fellegi, I. P. & Sunter, A. B. (1969) <i>A Theory for Record Linkage</i>	Foundational method for probabilistic record linkage
BAAI/bge-small-en-v1.5 model card	Sentence embedding model, 384 dimensions
ISO 15 — rolling bearings, boundary dimensions	Bearing designation to dimension mapping (6205 → 25×52×15 mm)
ASTM A105 / A106 / A193, API 600, IS 1364	Material and product standards cited in extracted specifications
Informatica, SAP MDG, Verdantis, PROSOL	Commercial master-data landscape reviewed for positioning

VERIFICATION DATE

All external facts in this document — UNSPSC class titles, the SAP MAKT-MAKTX field definition, and GeM and CPPP portal structure — were checked against their published sources on 31 August 2026.